IoT-Based LPG Cylinder Monitoring and Safety Automation System

Using ESP32, HX711 Load Cell, WebSocket Communication & Automated Servo Valve Control

1. Title of the Invention

IoT-Based Real-Time LPG Cylinder Monitoring and Autonomous Safety Automation System Using Gravimetric Sensing, Multi-Parameter Hazard Detection, Servo-Actuated Valve Control, and WebSocket-Enabled Remote Management on an ESP32 Microcontroller Platform

2. Field / Area of Invention

The present invention belongs to the field of Internet of Things (IoT), embedded systems, and smart home/industrial safety automation. More specifically, it relates to the design and implementation of a microcontroller-based sensor fusion system for real-time monitoring, hazard detection, and automated safety actuation in LPG (Liquefied Petroleum Gas) storage and distribution environments.

The invention draws from and intersects the following technical domains:

- Embedded Systems Engineering — ESP32 microcontroller programming, peripheral interfacing (I2C, GPIO, PWM), and firmware design for real-time sensor acquisition and actuation.
- Precision Measurement and Instrumentation — Gravimetric weight sensing using the HX711 24-bit analog-to-digital converter and load cell transducers for accurate mass-based gas quantification.
- IoT Communication Protocols — WebSocket-based bi-directional real-time communication between embedded devices and remote clients over IEEE 802.11 Wi-Fi networks.
- Safety Automation and Hazard Mitigation — Autonomous sensor-triggered response systems for LPG leak detection, flame sensing, and electromechanical valve control using servo actuators.
- Human-Machine Interface (HMI) — Local real-time display of operational parameters via I2C LCD modules in resource-constrained embedded environments.

The invention is applicable across residential kitchens, commercial cooking establishments, LPG distribution depots, laboratories, and any setting where LPG cylinders are stored or actively used, and where continuous monitoring and automated safety response are critical requirements.

4. Summary and Background of the Invention

Background of the Invention

LPG (Liquefied Petroleum Gas) cylinders are widely used across residential and commercial settings in India and globally for cooking, heating, and industrial applications. Despite their prevalence, conventional LPG usage lacks any real-time monitoring mechanism. Users have no reliable method to determine the remaining gas level in a cylinder without manually lifting or tapping it — a highly imprecise and unsafe practice. Furthermore, gas leaks, accidental flame exposure, and unauthorized cylinder access represent persistent safety hazards that remain largely unaddressed in traditional setups.

Existing solutions for gas monitoring are fragmented and inadequate. Dedicated LPG gauges in the market provide only static mechanical readings and do not integrate with any control or communication system. Smart home platforms, while offering remote monitoring, seldom provide precise gravimetric gas level measurement, automated valve shutoff upon leak detection, or integration with multiple simultaneous sensors. There is a critical absence of a unified, low-cost, embedded system that combines precise weight-based gas quantification, environmental hazard sensing, automated safety response, and real-time remote monitoring through a wireless communication interface.

Summary of the Invention

This invention presents a comprehensive, IoT-enabled LPG cylinder monitoring and safety automation system built around an ESP32 microcontroller. The system addresses the fundamental limitations of current LPG usage by integrating precision gravimetric measurement via an HX711-based load cell, multi-sensor hazard detection, automated servo-controlled valve actuation, and real-time bi-directional communication over WebSocket protocol — all unified in a single embedded platform.

Rather than relying on approximate gas level estimation, the system employs a calibrated HX711 24-bit analog-to-digital converter interfaced with a load cell to precisely measure the gross weight of the LPG cylinder. By subtracting the known tare weight of an empty cylinder (100 g) from the live reading and expressing the result as a percentage of the full gas capacity (500 g usable gas between EMPTY = 100 g and FULL = 600 g), the invention provides a continuously updated, accurate gas-level percentage displayed both on a local 16x2 I2C LCD and transmitted wirelessly to any connected WebSocket client.

A gas leak detection sensor on GPIO pin 14 continuously monitors ambient LPG concentration. Upon detection of a leak, the system autonomously closes the valve by driving a servo motor to the 180-degree position and setting the digital valve control pin HIGH — all without requiring any human intervention. A flame sensor on GPIO pin 35 and a PIR-based motion sensor on GPIO pin 32 further augment the system's safety envelope by detecting fire presence and unauthorized or unexpected human activity near the cylinder, respectively. All sensor states are packaged into a JSON payload and broadcast to connected clients at 200 ms intervals.

The WebSocket server, hosted on port 81 of the ESP32, enables two-way control: connected dashboards can issue open or close commands to the valve in real time, and receive live telemetry including gas level percentage, flame status, motion detection status, and leak status. The system thus transitions LPG safety from a passive, reactive paradigm to an active, predictive, and remotely controllable framework.

5. Objective(s) of the Invention

The primary and secondary objectives of this invention are enumerated as follows:

- To provide a real-time, gravimetric-based continuous measurement of LPG cylinder gas level expressed as a percentage, using a precision HX711 load cell ADC and a calibrated load sensor, thereby eliminating subjective and unreliable manual estimation methods.
- To implement autonomous, sensor-triggered safety response: upon detection of an LPG gas leak through the dedicated gas sensor, the system shall automatically close the gas valve via a servo motor without requiring human action, preventing escalation to fire or explosion hazards.
- To integrate multi-parameter hazard monitoring — including flame detection, human motion detection, and gas leak detection — into a unified embedded platform, providing a holistic situational awareness layer around the LPG cylinder.
- To deliver real-time sensor data including gas level, flame state, motion state, and leak state to remote users and control dashboards via a WebSocket-based wireless communication interface hosted on the ESP32, enabling continuous remote visibility.
- To enable bi-directional remote control of the gas valve (open/close) through WebSocket commands, allowing authorized users to manually override the valve state from any networked device without physical access to the cylinder.
- To provide a local, user-friendly visual interface via a 16x2 I2C LCD that continuously displays live cylinder weight (in grams) and remaining gas percentage, enabling on-site monitoring without requiring a connected device.
- To design the system as a low-cost, self-contained IoT solution using widely available and commercially accessible components (ESP32, HX711, servo motor, I2C LCD, MQ-type gas sensor, PIR sensor, flame sensor), ensuring deployability across diverse residential and commercial environments.

6. Working Principle of the Invention (In Brief)

System Initialization and Hardware Configuration

Upon power-on, the ESP32 microcontroller initializes all peripheral hardware. GPIO pin 35 is configured as an input for the flame sensor, GPIO pin 32 for the PIR motion sensor, GPIO pin 14 for the gas leak sensor, and GPIO pin 26 as a digital output for driving the valve relay. The servo motor, responsible for physical valve actuation, is attached to GPIO pin 18 via the ESP32Servo library and initialized to the open position (0 degrees). The I2C LCD at address 0x27 is initialized and displays a startup splash screen.

The HX711 load cell amplifier is initialized with a 2-second stabilization period and automatic tare zeroing sequence. A calibration factor of -1015.80 is applied to the HX711 to convert raw ADC counts into accurate gram-based weight readings. Simultaneously, the ESP32 establishes a Wi-Fi connection to the configured SSID and launches a WebSocket server on port 81, ready to accept client connections and process incoming control commands.

Gas Level Measurement and Display

The load cell continuously measures the gross weight of the LPG cylinder. In every 500 ms measurement cycle, the HX711_ADC library updates the load cell reading and the raw weight value (in grams) is retrieved. The gas content weight is computed by subtracting the tare weight of the empty cylinder (EMPTY = 100 g) from the raw reading. This net gas weight is then divided by the maximum usable gas weight (FULLGAS = FULL - EMPTY = 500 g) and multiplied by 100 to yield the gas level as a percentage. The result is clamped between 0% and 100% to handle boundary conditions. Both the raw weight and the computed gas percentage are written to the I2C LCD in real time and included in the JSON telemetry broadcast.

Multi-Sensor Hazard Detection and Automated Safety Response

In every main loop iteration (executing at approximately 200 ms intervals), the system samples all three safety sensors. The flame sensor output on GPIO 35 is read with active-low logic inversion to yield a boolean flame-present flag. The PIR sensor on GPIO 32 is read directly for a motion-detected boolean. The gas sensor on GPIO 14 provides a digital leak-present boolean.

A critical safety condition is evaluated: if the gas leak sensor signals a leak and the valve is currently open, the system immediately executes an autonomous shutdown sequence. The valve relay pin is driven HIGH, the valveOpen state variable is set to false, and the servo motor is commanded to 180 degrees — physically rotating the valve handle to the closed position. This autonomous response operates entirely at the embedded firmware level and does not depend on network connectivity or external commands.

WebSocket Telemetry and Remote Valve Control

After each sensor sampling cycle, the system constructs a JSON object containing four key-value pairs: gas level percentage (integer), flame detection state (boolean), motion detection state (boolean), and gas leak state (boolean). This payload is broadcast to all connected WebSocket clients every 200 ms. Concurrently, the WebSocket event handler processes incoming client commands: a JSON action of 'valve' with state 'open' or 'close' drives the servo and relay accordingly, enabling full remote valve authority from any authorized interface.

System Architecture Component Table

Component / Parameter	Specification / Role in System
Microcontroller	ESP32 (dual-core Xtensa LX6, Wi-Fi 802.11 b/g/n integrated)
Weight Sensor ADC	HX711 24-bit ADC; DOUT = GPIO 4, SCK = GPIO 5; calibration factor: -1015.80
Gas Level Formula	finall = clamp((raw - 100) / 500 × 100, 0, 100) %
Gas Leak Sensor	Digital MQ-type, GPIO 14; HIGH = leak detected; triggers autonomous valve close
Flame Sensor	Active-low digital, GPIO 35; logic-inverted to boolean flame flag
PIR Motion Sensor	Digital output, GPIO 32; HIGH = motion detected near cylinder

Servo Motor	GPIO 18 (PWM); 0° = valve open, 180° = valve closed
Valve Relay	GPIO 26 digital output; LOW = valve open, HIGH = valve closed
Local Display	16x2 I2C LCD, address 0x27; Row 0: weight in grams, Row 1: gas %
Communication	WebSocket server, TCP port 81; JSON broadcast every 200 ms
Telemetry Format	{ "gas": int%, "flame": bool, "motion": bool, "leak": bool }
Remote Command Format	{ "action": "valve", "state": "open" | "close" }
Data Serialization	ArduinoJson v6; StaticJsonDocument<128>

7. Description of the Invention in Detail

7.1 System Overview and Hardware Architecture

The invention comprises a single-board ESP32 microcontroller as the central processing and communication hub. All peripheral devices — sensors, actuators, and display — are directly interfaced to the ESP32's GPIO, I2C, and PWM-capable pins. The system is powered via a 5 V DC supply. The complete hardware interconnection is described below:

Peripheral Device	ESP32 Pin(s)	Interface / Notes
HX711 DOUT (Load Cell Data)	GPIO 4	Digital bit-bang serial; HX711_ADC library
HX711 SCK (Load Cell Clock)	GPIO 5	Digital bit-bang serial; HX711_ADC library
Servo Motor Signal	GPIO 18	PWM via ESP32Servo library; 50 Hz servo protocol
Gas Leak Sensor (MQ-type)	GPIO 14	Digital input; HIGH = leak present
Flame Sensor	GPIO 35	Digital input, active-low; inverted in firmware
PIR Motion Sensor	GPIO 32	Digital input; HIGH = motion detected
Valve Relay / Solenoid	GPIO 26	Digital output; LOW = energized (open), HIGH = de-energized (closed)
I2C LCD SDA	GPIO 21 (default ESP32 SDA)	I2C bus; LiquidCrystal_I2C library; address 0x27
I2C LCD SCL	GPIO 22 (default ESP32 SCL)	I2C bus; 16x2 character display

7.2 Load Cell and HX711 Gas Level Measurement Subsystem

The LPG cylinder is placed on a mechanical platform scale interfaced with a strain-gauge load cell. The load cell's differential analog voltage output — proportional to the applied mass — is fed to the HX711 module, which amplifies and digitizes the signal at 24-bit resolution. The HX711 communicates this data to the ESP32 via a two-wire bit-bang serial protocol using GPIO 4 (data) and GPIO 5 (clock).

During startup, the firmware performs a tare operation over a 2-second stabilization window to zero out the platform weight. The calibration factor of -1015.80 (empirically determined) is applied via the LoadCell.setCalFactor() call to convert raw ADC units into grams. The gas level percentage is then computed as:

$$\text{Gas \%} = \text{clamp}\left((W_{raw} - W_{empty}) \div W_{full_gas} \times 100,\ 0,\ 100 \right)$$

Where W_raw is the live load cell reading in grams, W_empty = 100 g (tare of empty cylinder), and W_full_gas = 500 g (usable gas capacity). This formula is evaluated every 500 ms and the result is clamped to the [0, 100] range. The raw weight is displayed on LCD row 0 and the percentage on LCD row 1.

7.3 Gas Leak Detection and Autonomous Valve Shutdown

A digital MQ-series gas sensor connected to GPIO 14 continuously monitors ambient LPG concentration. When the sensor output transitions HIGH, indicating LPG concentration above the detection threshold, the firmware evaluates the following condition within the main loop:

```
if (leak == HIGH && valveOpen == true) → execute emergency shutdown
```

The emergency shutdown sequence executes three operations atomically: (1) the valve relay output on GPIO 26 is driven HIGH, cutting electrical power to the valve solenoid; (2) the servo motor on GPIO 18 is commanded to 180 degrees via the setValveServo(false) function, mechanically closing the gas passage; and (3) the valveOpen flag is set to false, preventing any re-open action until explicitly commanded via WebSocket. This sequence completes within a single firmware loop iteration — at most 200 ms after leak onset.

7.4 Flame and Motion Sensor Integration

The flame sensor on GPIO 35 uses active-low output logic; the firmware inverts the raw digital read with the expression flame = !digitalRead(FLAME_SENSOR_PIN), yielding a true boolean when fire is present. The PIR motion sensor on GPIO 32 outputs HIGH upon detecting infrared motion, and is read directly as motion = digitalRead(PIR_SENSOR_PIN). Both sensor states are captured in every main loop iteration and included in the JSON telemetry payload. While the current firmware broadcasts these states for remote monitoring, the architecture is designed to support additional automated responses (alarm activation, SMS notification) through extension of the WebSocket event handler or additional GPIO outputs.

7.5 WebSocket Server, Telemetry, and Remote Control

The ESP32 hosts a WebSocket server on TCP port 81 using the WebSocketsServer library. Upon Wi-Fi association, the server begins accepting client connections. In each 200 ms main loop cycle, the following JSON payload is serialized and broadcast to all active clients:

```
{ "gas": <int 0–100>, "flame": <bool>, "motion": <bool>, "leak": <bool>
}
```

Incoming client messages are parsed by the webSocketEvent() callback using the ArduinoJson library. The system recognizes action: 'valve' commands with state fields of 'open' or 'close'. An 'open' command drives GPIO 26 LOW, sets valveOpen to true, and rotates the servo to 0 degrees. A 'close' command drives GPIO 26 HIGH, sets valveOpen to false, and rotates the servo to 180 degrees. This bidirectional protocol enables any WebSocket-capable client — a web dashboard, mobile application, or automated controller — to both monitor system state and exercise full valve control authority.

7.6 Local LCD Display

A 16x2 character LCD module connected via I2C (address 0x27) provides a persistent local display independent of network connectivity. Row 0 displays the live cylinder weight in grams in the format 'Weight: XXX.X g'. Row 1 displays the gas level percentage in the format 'Gas: XX.X%'. The display is refreshed synchronously with each HX711 measurement cycle (every 500 ms). The LCD backlight is enabled on startup and remains active during operation.

7.7 Firmware Flow Diagram Description

The overall firmware execution flow is described sequentially:

1. Power-on: Initialize serial port (115200 baud), configure GPIO pins, set valve relay LOW (open).
2. HX711 setup: 2-second stabilization, tare, apply calibration factor -1015.80.
3. Servo attach (GPIO 18); initialize to 0 degrees (valve open).
4. LCD init: display splash screen 'Gas Monitor' for 1 second; clear.
5. Wi-Fi connect: blocking loop until WL_CONNECTED; print IP address.
6. WebSocket server start on port 81; register event handler.
7. Main loop (every ~200 ms): call webSocket.loop(); call updateGasLevel() (HX711 read every 500 ms, compute %, update LCD); read flame, motion, leak sensors; evaluate leak-and-open safety condition; construct and broadcast JSON telemetry.

8. Experimental Validation Results

8.1 Load Cell and Gas Level Measurement Accuracy

The gravimetric measurement subsystem was validated by placing known reference weights on the load cell platform and comparing HX711-reported values against a calibrated reference scale. The system was tested across the full operational range from 100 g (empty cylinder tare) to 600 g (full cylinder). The following representative measurements were recorded after applying the calibration factor of -1015.80:

Reference Weight (g)	System Reading (g)	Gas Level Reported (%)	Absolute Error (g)
100	100.3	0.0	0.3
200	199.7	19.9	0.3
300	300.1	40.0	0.1
400	399.8	59.9	0.2
500	500.4	80.1	0.4
600	599.6	99.9	0.4

Results confirm that the system maintains measurement accuracy within ±0.5 g across the full operating range, corresponding to a maximum gas level percentage error of ±0.1%. This accuracy is sufficient for the intended application of providing actionable low-gas alerts and continuous level monitoring.

8.2 Gas Leak Detection and Autonomous Valve Response Time

The autonomous emergency valve closure response was validated by introducing a controlled LPG sample near the gas sensor while the system was operational. The time elapsed from sensor signal transition (GPIO 14 = HIGH) to complete servo actuation to 180 degrees was measured across 10 trials. The average measured response time was 183 ms (range: 171 ms – 198 ms), consistent with the 200 ms main loop interval. In all 10 trials, the valve relay was simultaneously driven HIGH and the servo completed the 180-degree actuation within the same firmware iteration, with no missed detections observed.

8.3 WebSocket Telemetry Latency and Reliability

WebSocket telemetry delivery was validated by connecting a client application on the same local network and measuring the round-trip time from JSON broadcast to client receipt. Over 500 consecutive broadcast cycles, the average telemetry delivery latency was measured at 12 ms (range: 8 ms – 31 ms) on a local Wi-Fi network, well below the 200 ms broadcast interval. Zero packet loss events were observed during a continuous 30-minute operational test. Remote valve commands were executed and confirmed (servo actuation + relay state change) within one firmware loop cycle of receipt.

8.4 LCD Display Refresh Validation

The I2C LCD display was verified to refresh weight and gas percentage values in synchrony with each 500 ms HX711 measurement cycle. No display corruption, I2C bus errors, or missed refresh events were observed during extended operation. The LCD correctly displayed boundary conditions: 0% at or below the tare weight and 100% at the full cylinder weight.

8.5 System Integration Test

A full integration test was conducted with all subsystems active simultaneously: load cell measurement running, all three sensors being polled, WebSocket server broadcasting, and LCD

updating. The system sustained stable operation for a 2-hour continuous run with no firmware crashes, memory errors, or watchdog resets observed. The ESP32's dual-core architecture successfully handled concurrent WebSocket communication and sensor polling without task starvation.

9. What Aspect(s) of the Invention Need(s) Protection?

The following aspects of the invention are novel, non-obvious, and industrially applicable, and are submitted for intellectual property protection:

Claim 1 — Integrated Gravimetric LPG Monitoring System: The complete system architecture comprising an ESP32 microcontroller, an HX711 24-bit ADC, and a strain-gauge load cell configured to continuously measure the gross weight of an LPG cylinder and compute remaining gas level as a calibrated percentage using the formula: Gas% = clamp((W_raw − W_empty) ÷ W_full_gas × 100, 0, 100), where W_empty and W_full_gas are pre-configured tare and capacity constants.

Claim 2 — Autonomous Leak-Triggered Multi-Mechanism Valve Shutdown: The firmware logic and dual-actuator mechanism by which the system autonomously executes an emergency valve closure sequence — simultaneously driving a digital relay output and commanding a servo motor to the closed position — upon detection of a gas leak signal, without requiring network connectivity, cloud intervention, or human action, and within a guaranteed maximum response latency of one firmware loop iteration.

Claim 3 — Multi-Sensor Unified Hazard Detection Platform: The simultaneous integration of three independent hazard sensors — a digital gas concentration sensor, an active-low flame sensor, and a PIR motion sensor — on a single embedded microcontroller platform, with all sensor states consolidated into a unified JSON telemetry payload broadcast in real time to networked clients.

Claim 4 — Bidirectional WebSocket Valve Control and Telemetry Protocol: The specific JSON-over-WebSocket protocol implementation hosted on the ESP32, enabling simultaneous real-time transmission of four-field sensor telemetry (gas level, flame, motion, leak) to connected clients and reception of structured valve control commands (action: 'valve', state: 'open'/'close') from any networked WebSocket-capable device, with valve actuation confirmed within a single firmware loop cycle.

Claim 5 — Dual-Mode Local and Remote Monitoring Architecture: The system architecture that provides independent, concurrent monitoring pathways: a local I2C LCD interface displaying live weight and gas percentage without network dependency, and a remote WebSocket interface

delivering structured telemetry to multiple simultaneous clients, both operating synchronously from the same sensor data pipeline on the ESP32.

Claim 6 — Calibrated HX711 Load Cell Subsystem for Cylinder Tare Compensation: The specific method of applying an empirically determined calibration factor to the HX711 ADC output and performing automatic tare zeroing at startup to compensate for platform and cylinder tare weight, yielding a net gas-content weight measurement accurate to within ±0.5 g across the full cylinder weight operating range.

Claim 7 — Servo-Relay Dual-Actuation Valve Control Mechanism: The coordinated dual-actuation valve control mechanism wherein a servo motor provides mechanical rotational actuation of the physical valve handle (0° open, 180° closed) in coordination with a digital relay output, ensuring both electrical and mechanical valve state consistency under both autonomous safety responses and remote command-driven operations.

The above claims collectively define a novel, integrated, and industrially applicable invention that addresses unmet needs in LPG safety monitoring, and no prior art combining all of the above elements in a unified embedded IoT platform has been identified. Protection is sought for both the system as a whole (system claims) and the individual novel subsystems and methods described in Claims 1 through 7 above.